lab 04

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Exercício 1

Carregue os pacotes readxl e tidyverse.

```r
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

Exercício 2

Crie uma variável chamada `fname` que contenha o caminho para o arquivo `WDIEXCEL.xlsx`, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando `file.exists`

```r
fname <- "WDIEXCEL.xlsx"

file.exists(fname)
```

```
[1] TRUE
```

Exercício 3

Leia a planilha "Country" e armazene no objeto countries.

```r
countries <- read_excel(fname, sheet = "Country")

head(countries)
```

```
# A tibble: 6 x 31
  `Country Code` `Short Name`             `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>                    <chr>        <chr>       <chr>
1 ABW            Aruba                    Aruba        Aruba       AW
2 AFE            Africa Eastern and Sou~ Africa East~ Africa Eas~ ZH
3 AFG            Afghanistan              Afghanistan  Islamic St~ AF
4 AFW            Africa Western and Cen~ Africa West~ Africa Wes~ ZI
5 AGO            Angola                   Angola       People's R~ AO
6 ALB            Albania                  Albania      Republic o~ AL
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Exercício 4

Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```r
paises <- c("BRA", "AFE", "CAN", "DMA", "ECU")

countries.5 <- countries %>%
  filter(`Country Code` %in% paises)

countries.5
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name`             `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>                    <chr>        <chr>       <chr>
1 AFE            Africa Eastern and Sou~ Africa East~ Africa Eas~ ZH
2 BRA            Brazil                   Brazil       Federative~ BR
```

```
3 CAN            Canada                 Canada       Canada      CA
4 DMA            Dominica               Dominica     Commonweal~ DM
5 ECU            Ecuador                Ecuador      Republic o~ EC
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Exercíco 5

Leia a planilha "Data" e armazene no objeto WDI.

```r
WDI <- read_excel(fname, sheet = "Data")

head(WDI)
```

```
# A tibble: 6 x 70
  `Country Name`  `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
  <chr>           <chr>          <chr>            <chr>             <dbl>  <dbl>
1 Africa Eastern~ AFE            Access to clean~ EG.CFT.ACCS.ZS       NA     NA
2 Africa Eastern~ AFE            Access to clean~ EG.CFT.ACCS.RU.~     NA     NA
3 Africa Eastern~ AFE            Access to clean~ EG.CFT.ACCS.UR.~     NA     NA
4 Africa Eastern~ AFE            Access to elect~ EG.ELC.ACCS.ZS       NA     NA
5 Africa Eastern~ AFE            Access to elect~ EG.ELC.ACCS.RU.~     NA     NA
6 Africa Eastern~ AFE            Access to elect~ EG.ELC.ACCS.UR.~     NA     NA
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>,
#   `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>, ...
```

Exercício 6

Selecione apenas as colunas: - Country Code - Indicator Code - E todos os anos de 1960-2018

```r
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)

head(WDI)
```

```
# A tibble: 6 x 61
  `Country Code` `Indicator Code`  `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>              <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
1 AFE            EG.CFT.ACCS.ZS        NA     NA     NA     NA     NA     NA
2 AFE            EG.CFT.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
3 AFE            EG.CFT.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
4 AFE            EG.ELC.ACCS.ZS        NA     NA     NA     NA     NA     NA
5 AFE            EG.ELC.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
6 AFE            EG.ELC.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>,
#   `1995` <dbl>, `1996` <dbl>, `1997` <dbl>, `1998` <dbl>, `1999` <dbl>, ...
```

Exercício 7

Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```r
WDI <- pivot_longer(WDI, cols = `1960`:`2018`, names_to = "YEAR", values_to = "VALUE")

WDI
```

```
# A tibble: 23,421,230 x 4
   `Country Code` `Indicator Code` YEAR  VALUE
   <chr>          <chr>            <chr> <dbl>
 1 AFE            EG.CFT.ACCS.ZS   1960     NA
 2 AFE            EG.CFT.ACCS.ZS   1961     NA
 3 AFE            EG.CFT.ACCS.ZS   1962     NA
 4 AFE            EG.CFT.ACCS.ZS   1963     NA
 5 AFE            EG.CFT.ACCS.ZS   1964     NA
 6 AFE            EG.CFT.ACCS.ZS   1965     NA
 7 AFE            EG.CFT.ACCS.ZS   1966     NA
```

```
 8 AFE            EG.CFT.ACCS.ZS   1967     NA
 9 AFE            EG.CFT.ACCS.ZS   1968     NA
10 AFE            EG.CFT.ACCS.ZS   1969     NA
# i 23,421,220 more rows
```

Exercício 8

Confirme se a coluna referente ao ano está em formato inteiro

```r
typeof(WDI$YEAR)
```

```
[1] "character"
```

```r
WDI$YEAR <- as.integer(WDI$YEAR)
typeof(WDI$YEAR)
```

```
[1] "integer"
```

Exercício 9

Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```r
WDI <- WDI %>%
  left_join(
    countries.5 %>%
      select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) %>%
  filter(
    !is.na(`Short Name`),
    `Indicator Code` == "SP.DYN.CBRT.IN"
  ) %>%
  select(
    `Country Code`,
```

```
    `Short Name`,
    YEAR,
    VALUE
  )

WDI
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`                  YEAR VALUE
   <chr>          <chr>                        <int> <dbl>
 1 AFE            Africa Eastern and Southern  1960  47.7
 2 AFE            Africa Eastern and Southern  1961  47.7
 3 AFE            Africa Eastern and Southern  1962  47.8
 4 AFE            Africa Eastern and Southern  1963  47.8
 5 AFE            Africa Eastern and Southern  1964  47.8
 6 AFE            Africa Eastern and Southern  1965  47.8
 7 AFE            Africa Eastern and Southern  1966  47.9
 8 AFE            Africa Eastern and Southern  1967  47.9
 9 AFE            Africa Eastern and Southern  1968  47.9
10 AFE            Africa Eastern and Southern  1969  47.8
# i 285 more rows
```

Exercício 10

Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```
WDI <- arrange(WDI, `Short Name`, YEAR)

WDI
```

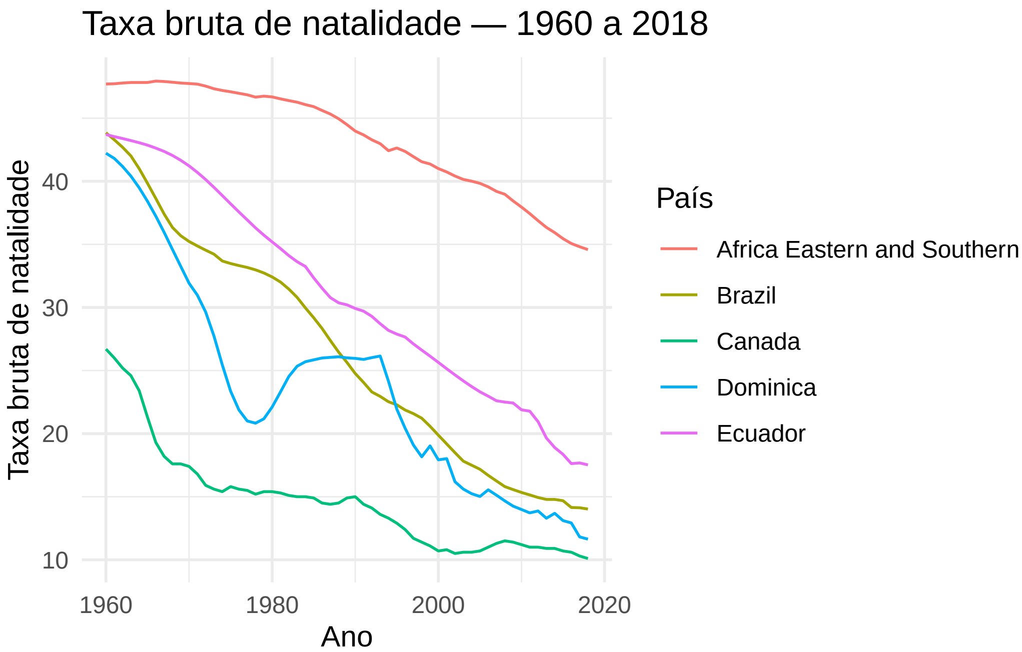
```
# A tibble: 295 x 4
   `Country Code` `Short Name`                  YEAR VALUE
   <chr>          <chr>                        <int> <dbl>
 1 AFE            Africa Eastern and Southern  1960  47.7
 2 AFE            Africa Eastern and Southern  1961  47.7
 3 AFE            Africa Eastern and Southern  1962  47.8
 4 AFE            Africa Eastern and Southern  1963  47.8
 5 AFE            Africa Eastern and Southern  1964  47.8
 6 AFE            Africa Eastern and Southern  1965  47.8
 7 AFE            Africa Eastern and Southern  1966  47.9
 8 AFE            Africa Eastern and Southern  1967  47.9
```

```
 9 AFE            Africa Eastern and Southern  1968  47.9
10 AFE            Africa Eastern and Southern  1969  47.8
# i 285 more rows
```

Exercício 11

Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```r
ggplot(
  WDI,
  aes(
    x = YEAR,
    y = VALUE,
    color = `Short Name`,
    group = `Short Name`
  )
) +
  geom_line(na.rm = TRUE) +
  labs(
    x = "Ano",
    y = "Taxa bruta de natalidade",
    color = "País",
    title = "Taxa bruta de natalidade - 1960 a 2018"
  ) +
  theme_minimal()
```




```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-09 14:08:04 -03"
```